

Panel MS-AR(1): common trend vs Pareto random intercepts

Quarterly dummies, real GDP per worker

1 Model

Joint panel Markov-switching AR(1) around a linear trend with common quarterly dummies (Q1 omitted):

$$y_{it} = a_i + g t + d_2 Q2_t + d_3 Q3_t + d_4 Q4_t + z_{it}, \quad (1)$$

$$z_{i,t+1} = (1 - \rho(s_{it}))\mu(s_{it}) + \rho(s_{it}) z_{it} + \sigma(s_{it}) \varepsilon_{it}, \quad (2)$$

$$s_{i,t+1} \sim \Pi(\cdot | s_{it}). \quad (3)$$

Specification 1 sets $a_i \equiv a$. Specification 2 draws a_i from a shifted Type II Pareto (Lomax) on $[m, \infty)$, with location m , scale σ , and shape α : $P(a_i > x) = (1 + (x - m)/\sigma)^{-\alpha}$ for $x \geq m$. The country likelihood integrates a_i by 15-point Gauss–Legendre on the probability-integral-transform scale. The slope g and the quarterly dummies are common. $|\rho| < 0.99$. Median μ pinned at 0. Cycles use posterior-mean a_i . Standard errors are delta-method from a numerical Hessian.

2 Common intercept, linear trend, and quarterly dummies

2026-09-14 panel, full sample: log real GDP per worker. 90 countries, 8489 observations, 1963-Q1–2026-Q2. Fitted countries: 90. Observations: 8489. Log-likelihood: 13182.75. Ergodic mean of the cycle $E[z] = 0.5738$ (0.0863). Common intercept $a = -5.4283$ (0.1150). Common slope $g = -0.0004$ (0.0016). Seasonal dummies (Q1 omitted): $d_{Q2} = 0.0161$ (0.0008), $d_{Q3} = 0.0106$ (0.0009), $d_{Q4} = 0.0432$ (0.0008).

Table 1: Regime AR(1), transition matrix, and ergodic π . Columns are regimes.

	(0)	(1)	(2)
μ	-0.8032 (0.3159)	0.0000 —	1.4857 (0.0734)
σ	0.1789 (0.0049)	0.0660 (0.0012)	0.0281 (0.0004)
ρ	0.9900	0.9900 (0.0000)	0.9900 (0.0000)
Π_0	0.9786 (0.0060)	0.0171 (0.0050)	0.0042 (0.0030)
Π_1	0.0040 (0.0013)	0.9812 (0.0028)	0.0148 (0.0025)
Π_2	0.0016 (0.0012)	0.0138 (0.0023)	0.9846 (0.0021)
π	0.1159 (0.0248)	0.4351 (0.0346)	0.4490 (0.0362)

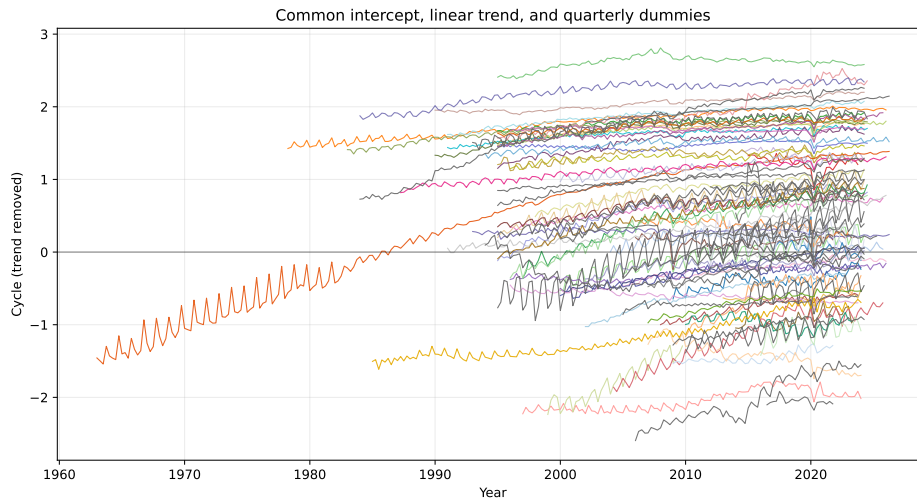


Figure 1: Country cycles after removing the linear trend and quarterly means.

3 Pareto RE intercepts, common trend and quarterly dummies

2026-09-14 panel, full sample: log real GDP per worker. 90 countries, 8489 observations, 1963-Q1–2026-Q2. Fitted countries: 90. Observations: 8489. Log-likelihood: 13439.80. Ergodic mean of the cycle $E[z] = 0.1664$ (0.0756). Random intercepts Type II Pareto on $[m, \infty)$ with location $m = -7.4177$ (0.1116), scale = 16.4022 (6.2334), shape $\alpha = 19.5016$ (6.8641). Posterior-mean a_i : mean -6.145, min -7.393, max -4.231. Common slope $g = 0.0167$ (0.0012). Seasonal dummies (Q1 omitted): $d_{Q2} = 0.0164$ (0.0008), $d_{Q3} = 0.0107$ (0.0009), $d_{Q4} = 0.0423$ (0.0008).

Table 2: Regime AR(1), transition matrix, and ergodic π . Columns are regimes.

	(0)	(1)	(2)
μ	-0.2856 (0.1012)	0.0000 —	0.6033 (0.1080)
σ	0.1678 (0.0047)	0.0653 (0.0011)	0.0266 (0.0004)
ρ	0.6946 (0.0339)	0.9900 (0.0001)	0.9900 (0.0000)
Π_0	0.9833 (0.0048)	0.0083 (0.0038)	0.0083 (0.0038)
Π_1	0.0029 (0.0009)	0.9837 (0.0026)	0.0134 (0.0025)
Π_2	0.0014 (0.0007)	0.0126 (0.0020)	0.9860 (0.0020)
π	0.1128 (0.0256)	0.4188 (0.0366)	0.4684 (0.0373)

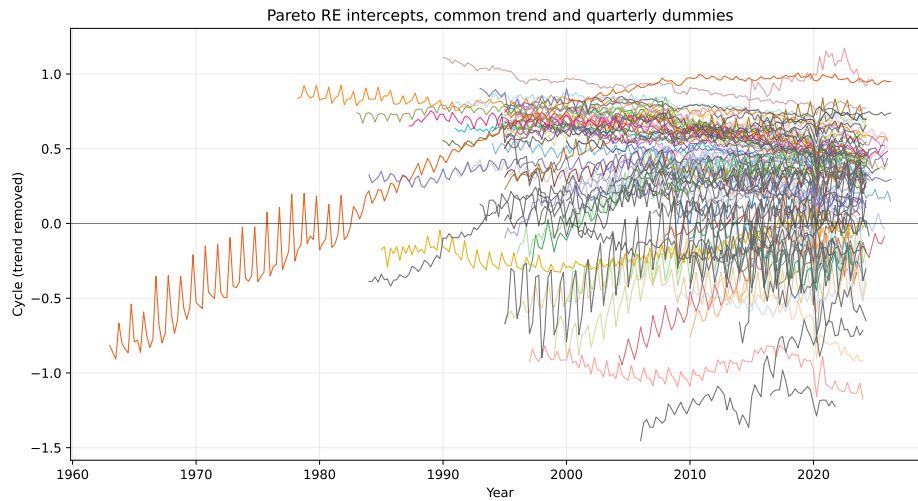


Figure 2: Country cycles after removing the linear trend and quarterly means.